Graph Variational Latent Space (GVLS) for Quantum Graph Neural Networks
Mid-Term Internship Report — Mizzou QIC Summer 2026

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1. Research Hypothesis and Objective

Quantum Graph Neural Networks (QGNNs) may offer advantages for graph-structured problems, but current NISQ hardware cannot directly process large graphs such as citation networks, molecular graphs, and knowledge bases. This project tests whether a **graph-structured variational latent space** can compress large graphs into compact representations suitable for QGNN input while preserving useful relational structure. The proposed **Graph Variational Latent Space (GVLS)** model maps an input graph with N nodes into a learned sparse latent graph with adjacency A_z and node embeddings $\tilde{z}$. A secondary contribution is GVLS as a standalone classical graph representation learning method competitive with recent variational graph autoencoders.

2. Progress on Quantum Methods Development

GVLS is a hybrid classical-quantum pipeline. The implemented classical component performs graph compression; the quantum component will consume the compressed graph. The encoder is a two-layer GCN that maps node features and adjacency into Gaussian posteriors $(\mu, \log \sigma^2)$ with latent dimension $d \ll N$. A differentiable **Latent Graph Learner** builds a sparse latent adjacency A_z using attention, cosine Feature Graph Parameterization (FGP), or an MLP while keeping top-k neighbors per node. Message passing over A_z refines embeddings into $\tilde{z}$, and the decoder reconstructs adjacency as $\hat{A} = \sigma(\tilde{z}\tilde{z}^{\mathsf{T}})$.

*The key novelty is that GVLS learns a **graph-structured latent space** rather than independent flat latent vectors.* The compressed graph $(\tilde{z}, A_z)$ is the planned input to a parameterized quantum circuit-based GNN, so qubit requirements can scale with latent dimension or compressed node count rather than the original graph size.

3. Computing Experimental Design

The main validation task is **link prediction** on citation benchmarks: Cora (2,708 nodes, 5,429 edges), CiteSeer (3,327 nodes, 4,732 edges), and PubMed (19,717 nodes, 44,338 edges) [1], following [2] protocol. Secondary evaluations include node classification and graph compression quality using reconstruction F1, bits-per-edge, and rate-distortion curves across d. The trained GVLS encoder will serve as classical preprocessing; the compression stage itself does not require quantum training.

4. Quantum Computing Requirements

Resource	Requirement
Qubit count	Scales with latent dimension d or compressed node count; currently $d = 128$, with target $d \leq 32$ for near-term NISQ experiments.
Gate depth	Expected to remain shallow, determined by 1–2 QGNN layers.
Classical preprocessing	Full GVLS forward pass on CPU/GPU: encoder, latent graph inference, and latent message passing.
Hybrid architecture	Classical GVLS compression $\rightarrow$ QGNN inference on $(\tilde{z}, A_z)$.

GVLS is useful for quantum computing because it can reduce a PubMed-scale graph with nearly 20,000 nodes into a much smaller latent graph. Top-k sparsification also limits edge density, reducing the entanglement structure required by the QGNN.

5. Preliminary Results

NAS was completed for all three datasets. The best Cora NAS validation AUC reached **0.9438**. Full training with NAS-selected configurations produced the following held-out link prediction AUC-ROC results.

Dataset	Split	GAE	LGAE	GNAE	VGNAE	GVLS
Cora	20%	0.782	0.866	0.887	0.890	**0.870**
Cora	80%	0.922	0.938	0.956	0.954	**0.917**
CiteSeer	20%	0.786	0.906	0.946	0.941	**0.941**
CiteSeer	80%	0.894	0.955	0.965	0.970	**0.929**
PubMed	20%	0.937	0.946	0.950	0.951	**0.835**
PubMed	80%	0.967	0.974	0.975	0.976	**0.934**

GVLS is competitive on Cora and CiteSeer, including parity with VGNAE on CiteSeer at the 20% split. PubMed performance is weaker in the 20% low-data regime. Importantly, GVLS learns a non-trivial latent graph rather than copying the input graph; for example, the Cora latent edge density is approximately 0.007. These results support further study of GVLS both as a quantum preprocessing method and as a classical graph learning contribution.

6. Planned Future Work

The next phase will turn validation results into a compression pipeline and quantum proof of concept. Immediate tasks are: (1) evaluate graph compression with reconstruction F1, bits-per-edge, and rate-distortion curves; (2) improve PubMed through ablations over graph method, prior, and β; (3) test node classification using a linear probe on $\tilde{z}$; (4) integrate $(\tilde{z}, A_z)$ with a parameterized QGNN; and (5) visualize latent graph structure using A_z comparisons and t-SNE/UMAP plots. Deliverables include ablation tables, paper sections, poster/report materials, and code release. GVLS may be submitted to graph learning venues such as NeurIPS, ICLR, or CIKM; the quantum component can later target QML venues or workshops.

7. Reflections

Several implementation choices were critical. Replacing mutual top-k intersection with **union symmetrization** prevented the latent graph from becoming empty at realistic graph sizes and preserved gradient flow. Overall, achieving VGNAE-level performance on CiteSeer while simultaneously learning a compressed latent graph is a strong signal that the GVLS idea is sound. The remaining work is to quantify compression quality and demonstrate QGNN integration.

References

[1] Z. Yang, W. W. Cohen, and R. Salakhutdinov, “Revisiting semi-supervised learning with graph embeddings,” in *Proceedings of the 33rd International Conference on Machine Learning*, 2016.

[2] S. J. Ahn and M. Kim, “Variational graph normalized autoencoders,” in *Proceedings of the 30th ACM international conference on information & knowledge management*, pp. 2827–2831, 2021.